

Do Factor Premia Travel?

Machine-Learning Factor Timing in Emerging-Market Equities

Yang Xiao*

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Preliminary results; please do not cite.

Abstract

I ask whether the time-varying predictability of equity factor returns documented in developed markets is present in emerging markets, and whether forecasting models trained only on developed-market history carry over. Using the Jensen et al. [2023] global factor returns for 153 characteristics across 23 developed and 24 emerging markets, I forecast each country-factor return one month ahead with pooled linear, regularized, tree and neural-network models, refitted annually out of sample. I first show, on known-truth simulations, that the standard benchmark for this exercise is inadequate in a multi-country panel: because each country-factor mean is estimated from a short and noisy history, a model can beat it by pooling information across countries, with no timing ability at all. Against the own historical mean, a nominal 5% Clark–West test rejects in 87.5% of simulated panels with no predictability; against a cross-country shrinkage mean it rejects in 5.0%, with 92.5–100% power when predictability exists. I therefore evaluate timing against the shrinkage benchmark and measure economic value through timing portfolios net of trading costs. In 801,536 emerging-market country-factor-months from 2000 to 2025, the shrinkage mean alone beats the own historical mean by more than any learned model does (R_{OS}^2 of 1.20%), so the gains the standard benchmark would credit to timing are gains from pooling. On capped value-weighted factors, every learned model has a negative R_{OS}^2 against the shrinkage mean (−0.05% to −0.48%), although all twelve Clark–West statistics are positive and ten exceed the 5% critical value: the forecasts point the right way but are two and a half to five times too large. Models trained only on developed markets carry at least as much directional information about emerging-market factor returns as models trained on emerging markets. Net of 20 basis points per unit of turnover, no learned-model timing portfolio beats a static equal-weight factor portfolio, and the highest net Sharpe ratio comes from a slow-moving tilt toward factors with high shrunk historical means. Equal-weighted factors, which give small stocks more weight, are the exception: most learned models beat the shrinkage mean and their timing portfolios beat the static portfolio, though not the shrinkage-mean tilt. Two exploratory extensions leave these conclusions intact: shrinking forecasts toward the benchmark in real time

*The Graduate Center, City University of New York. Email: yxiao1@gc.cuny.edu. Code: <https://github.com/YangXiaoShawn>.

lets emerging-market-trained timing portfolios beat the static portfolio but not the tilt, and U.S. state variables make every model worse.

JEL: G11, G12, G15, C53. **Keywords:** factor timing, emerging markets, machine learning, out-of-sample evaluation, shrinkage.

1 Introduction

Equity factor premia — the return spreads associated with value, momentum, profitability, low risk and many other characteristics — are among the most studied objects in empirical finance. Much of that evidence comes from the United States, and a long-standing concern is that part of it reflects data mining or disappears once published [McLean and Pontiff, 2016]. Jensen et al. [2023] construct 153 factors consistently across 93 countries and argue that most of them replicate and that factor premia are present across many markets. Whether factor returns are also *predictable over time*, and whether that predictability generalizes across markets, is a separate and less settled question.

Emerging markets are a natural place to ask it. They differ from developed markets in liquidity, trading costs, the mix of retail and institutional investors, and exposure to global capital flows. If time variation in factor premia reflects slow-moving compensation for risk that is priced globally, a model that learns it in developed markets should transfer. If instead it reflects local frictions or local investor behavior, predictability learned elsewhere should not carry over, and emerging markets may display patterns of their own.

I study both possibilities in a single out-of-sample design. For each country-factor pair, seven point-in-time predictors forecast next month’s factor return. I estimate pooled linear, elastic-net, gradient-boosted-tree and neural-network models, refitted every January on labels already realized, under three training scopes: emerging markets only, developed markets only (the transfer test), and both pooled.

The first contribution is methodological. The standard out-of-sample benchmark for a return forecast is the historical mean [Welch and Goyal, 2008, Campbell and Thompson, 2008]. In a panel of thousands of short country-factor series, each historical mean is itself estimated with large error, and any model that shares information across series can beat it simply by shrinking noisy means toward their cross-sectional center [Efron and Morris, 1975]. That is an improvement in estimating the *unconditional* premium, not evidence of timing. On simulated panels with a known data-generating process (Appendix A), a nominal 5% Clark–West test against the own historical mean rejects in 87.5% of draws that contain no predictability whatsoever. Replacing the benchmark with a cross-country shrinkage mean, and training models to forecast deviations from it, restores the correct size (5.0%) while keeping power at 92.5–100%. All timing claims in the paper are made against this benchmark.

The second contribution is the transfer test itself: holding the emerging-market test set fixed, I compare models trained on emerging-market history with models trained only on developed-market history. The third is economic: I translate forecasts into within-country factor-timing portfolios and evaluate them net of the costs of the timing trades, which matters because emerging-market factor legs are thinner and more expensive to trade [Novy-Marx and Velikov, 2016].

Four findings emerge from 801,536 emerging-market country-factor-month forecasts over 2000–

2025. First, the benchmark problem is large in the data. Against the own historical mean every model posts an R_{OS}^2 near 1% with Clark–West statistics above 8, but the shrinkage mean alone does better (1.20%, $t = 9.05$). Against the shrinkage mean, all twelve model-by-scope combinations have negative R_{OS}^2 , between -0.05% and -0.48% . Their Clark–West statistics are nonetheless positive, ten of twelve exceed the one-sided 5% critical value, and those of gradient-boosted trees exceed 3.6. Regressing realized on forecast deviations from the benchmark gives slopes of 0.19–0.40: the forecasts contain information about time variation but overstate it.

Second, developed-market training transfers in a directional sense. Holding the emerging-market test set fixed, models trained only on developed markets have Clark–West statistics at least as large as those of models trained on emerging markets, but their forecasts are more dispersed and their R_{OS}^2 lower.

Third, the economic value of timing does not survive trading costs. Learned-model portfolios raise gross mean returns from 2.4% to 3.1–3.8% a year, but with monthly turnover of 0.41–0.63 their net Sharpe ratios at 20 basis points (0.71–1.26) fall below the static factor portfolio’s 1.48. The highest net Sharpe ratio, 1.91, belongs to a portfolio that tilts toward factors with high shrunk historical means and barely trades.

Fourth, factor weighting matters. With equal-weighted factors, which give small stocks more weight, most learned models beat the shrinkage mean (R_{OS}^2 up to 0.17%) and every learned-model timing portfolio beats the static portfolio net of costs, although none beats the shrinkage-mean tilt. With uncapped value-weighted factors the evidence is weaker than in the baseline. This is consistent with predictability concentrated in small stocks, where trading is most expensive; the cost assumption here does not vary with factor weighting.

Two exploratory extensions, chosen after seeing these results, do not change them. Scaling each forecast toward the benchmark by a slope estimated inside the training window cuts turnover by a third to a half and lifts the emerging-market-trained portfolios above the static portfolio net of costs, but leaves every R_{OS}^2 on capped value-weighted factors negative and every portfolio below the shrinkage-mean tilt. Adding U.S. state variables (VIX, the term spread and the dollar) makes every model worse.

2 Related literature

Factors across countries. Rouwenhorst [1999] documents that return factors in emerging markets resemble those in developed markets, and Asness et al. [2013] show that value and momentum premia appear across markets and asset classes. Jacobs and Müller [2020] study whether the post-publication decline in anomaly returns documented for the United States extends internationally. Jensen et al. [2023] provide the consistently constructed global data used here.

Factor timing. Haddad et al. [2020] show that the dominant components of factor returns are

predictable from their valuations. Gupta and Kelly [2019] and Ehsani and Linnainmaa [2022] document factor momentum: factor returns are positively autocorrelated. Own trailing means therefore enter my predictor set, and a pure factor-momentum rule serves as a benchmark strategy.

Machine learning and return prediction. Gu et al. [2020] show that tree ensembles and neural networks improve U.S. return forecasts relative to linear models; Kelly et al. [2024] argue that complex models can improve out-of-sample prediction; Leippold et al. [2022] apply machine learning to the Chinese cross-section. My setting forecasts factor portfolios rather than individual stocks, which averages away idiosyncratic noise and makes the premia directly tradable.

Out-of-sample evaluation. Welch and Goyal [2008] show that many predictors fail to beat the historical mean out of sample; Campbell and Thompson [2008] introduce the out-of-sample R^2 used here; Clark and West [2007] provide the test for nested forecasts. My contribution is to show that in a multi-series panel the historical-mean benchmark must itself be pooled for these tests to measure timing.

3 Data

I use the monthly factor returns of Jensen et al. [2023], downloaded from <https://jkpfactors.com>. Each factor is a long-short portfolio sorted on one of 153 characteristics, signed so that the long leg is the side the original study predicts to earn higher returns, and capped value-weighted. Returns are in U.S. dollars in excess of the risk-free rate. I classify 23 countries as developed and 24 as emerging using the current MSCI market classification and drop country-factor-months in which either leg holds fewer than five stocks.

The file used here was downloaded from jkpfactors.com in September 2026 and runs through December 2025. Table 1 reports coverage. Most developed markets start in 1986 and the United States in 1926; emerging markets start between 1986 (South Africa) and 2005 (Peru). After the five-stock filter the Czech Republic (42 months, 1995–2000) and Hungary (5 months) are too thin to meet the predictors’ history requirements and contribute no out-of-sample observations, so the emerging-market test set covers 22 markets.

4 Empirical design

4.1 Target and predictors

Let $i = (c, f)$ index a country-factor pair and $r_{i,t}$ its return in month t . The target is $r_{i,t+1}$. Predictors observed at the end of month t are the own return $r_{i,t}$; own trailing means over 3, 12 and 36 months; own 12-month volatility; the average 12-month return of all factors in country c ; and the average 12-month return of factor f across developed markets. Every predictor uses returns

Table 1: Coverage of the JKP factor returns by market

Market	Factors	First	Last	Obs.	Market	Factors	First	Last	Obs.
<i>Developed markets (23)</i>									
Australia	153	1986-01	2025-12	62,115	Japan	153	1986-01	2025-12	66,498
Austria	153	1986-09	2025-12	55,750	Netherlands	153	1986-01	2025-12	61,889
Belgium	153	1986-01	2025-12	59,526	New Zealand	150	1986-03	2025-12	52,179
Canada	153	1982-03	2025-12	77,866	Norway	153	1986-01	2025-12	59,032
Denmark	153	1986-01	2025-12	60,340	Portugal	150	1989-09	2025-12	45,885
Finland	153	1987-01	2025-12	58,253	Singapore	153	1986-01	2025-12	60,133
France	153	1986-01	2025-12	62,761	Spain	153	1986-01	2025-12	60,355
Germany	153	1986-01	2025-12	62,658	Sweden	153	1986-01	2025-12	61,235
Hong Kong	153	1986-01	2025-12	61,797	Switzerland	153	1986-01	2025-12	61,225
Ireland	148	1989-02	2025-12	45,405	United Kingdom	153	1986-01	2025-12	68,945
Israel	153	1995-02	2025-12	47,873	United States	153	1926-01	2025-12	146,430
Italy	153	1986-01	2025-12	60,808					
<i>Emerging markets (24)</i>									
Brazil	153	1994-09	2025-12	41,874	Malaysia	153	1987-03	2025-12	59,338
Chile	152	1989-01	2025-12	52,377	Mexico	150	1987-09	2025-12	54,509
China	153	1993-11	2025-12	50,042	Peru	142	2005-05	2025-12	29,905
Colombia	145	1993-02	2025-12	29,043	Philippines	153	1993-07	2025-12	44,601
Czech Republic	35	1995-09	2000-02	849	Poland	152	1994-03	2025-12	41,792
Egypt	144	1997-04	2025-12	29,784	Qatar	143	2003-11	2025-12	29,382
Greece	152	1989-10	2025-12	44,765	Saudi Arabia	152	2001-11	2025-12	38,542
Hungary	20	2000-09	2001-01	41	South Africa	153	1986-01	2025-12	61,246
India	153	1988-09	2025-12	48,686	Taiwan	153	1988-02	2025-12	53,457
Indonesia	153	1990-03	2025-12	53,298	Thailand	152	1988-07	2025-12	54,436
Korea	153	1988-07	2025-12	53,855	Turkey	153	1990-04	2025-12	49,259
Kuwait	147	2001-04	2025-12	34,382	United Arab Emirates	144	2003-03	2025-12	31,468

Notes: Monthly capped value-weighted factor returns of [Jensen et al. \[2023\]](#), downloaded from [jkpfactors.com](#). Country-factor-months in which a leg holds fewer than 5 stocks are dropped. “Obs.” counts country-factor-months after that filter. Markets are classified by the current MSCI classification.

through month t only; a perturbation test confirms that altering returns after t leaves every month- t predictor unchanged.

4.2 Benchmarks

The own historical mean is $\bar{r}_{i,t} = n_{i,t}^{-1} \sum_{s \leq t} r_{i,s}$, where $n_{i,t}$ counts observed months. The shrinkage mean pulls it toward $\bar{r}_{g,f,t}$, the expanding mean of factor f pooled across all countries in the same region g :

$$\tilde{r}_{i,t} = \frac{n_{i,t} \bar{r}_{i,t} + \kappa \bar{r}_{g,f,t}}{n_{i,t} + \kappa}, \quad \kappa = 120. \quad (1)$$

Short histories are shrunk heavily; long histories retain their own mean. I report results against both benchmarks and use $\kappa \in \{60, 240\}$ as robustness.

4.3 Models and out-of-sample protocol

Learned models forecast the deviation $r_{i,t+1} - \tilde{r}_{i,t}$, and the forecast is $\hat{r}_{i,t+1} = \tilde{r}_{i,t} + \hat{d}_{i,t}$. Models are pooled OLS, elastic net, gradient-boosted regression trees, and a three-seed ensemble of two-layer

neural networks. The trees and the neural network are fitted on random subsamples of at most 400,000 and 133,333 training rows; the neural network’s early stopping uses a random validation split within the training window. Features are standardized and training targets winsorized at the 0.5 and 99.5 percentiles using the training window only; hyperparameters are fixed in advance. Two rules without estimation complete the set: the shrinkage mean itself and factor momentum (the own 12-month mean).

Every January of year Y , models are refitted on all observations whose target month precedes January of Y and forecast every target month in Y . Out-of-sample evaluation begins in 2000. For emerging-market test observations I compare three training scopes: *own* (emerging markets), *DM-only* (developed markets) and *pooled*.

4.4 Statistical evaluation

For forecasts $\hat{r}$ and benchmark b ,

$$R_{OS}^2 = 1 - \frac{\sum_{i,t} (r_{i,t+1} - \hat{r}_{i,t+1})^2}{\sum_{i,t} (r_{i,t+1} - b_{i,t})^2}. \quad (2)$$

The Clark–West statistic uses $f_{i,t+1} = 2(r_{i,t+1} - b_{i,t})(\hat{r}_{i,t+1} - b_{i,t})$. Because forecast errors are correlated across country-factor pairs within a month, I average f across the cross-section each month and test the mean of the resulting monthly series with [Newey and West \[1987\]](#) standard errors (three lags).

4.5 Economic evaluation

Within each country c and month t , the timing portfolio holds factor f with weight $w_{c,f,t} = \hat{r}_{c,f,t+1} / \sum_{f'} |\hat{r}_{c,f',t+1}|$, so gross exposure is one. The static benchmark holds every factor with weight $1/N_{c,t}$. Country portfolios are averaged with equal weights. Turnover is $\sum_f |w_{c,f,t} - w_{c,f,t-1}|$, and net returns deduct a cost per unit of turnover (20 basis points in the baseline; 10 and 40 as robustness). The cost applies only to timing trades, since the factor returns are already gross of their own rebalancing.

5 Results

5.1 Is there predictability beyond the unconditional premium?

Table 2 reports pooled out-of-sample results for the emerging-market test set: 801,536 country-factor-months in 22 markets, 2000–2025.

Against the own historical mean the evidence looks strong. Every learned model has an R_{OS}^2 between 0.73% and 1.16%, with Clark–West statistics between 8.3 and 9.0. But the shrinkage mean, which forecasts no time variation at all, does better than any of them (1.20%, $t = 9.05$).

This is the pattern the simulations in Appendix A warn about: the improvement over the own mean comes from pooling noisy country-factor means, not from timing. Factor momentum, which uses the trailing 12-month mean as its forecast, has an R_{OS}^2 of -6.2% .

Against the shrinkage mean, every learned model has a negative R_{OS}^2 , from -0.05% (elastic net, EM-trained) to -0.48% (neural network, DM-trained). The Clark–West statistics are nonetheless positive in all twelve cases and above the one-sided 5% critical value of 1.645 in ten. For the linear models they range from 1.56 to 2.10; for gradient-boosted trees from 3.66 to 4.25, above 2.64, the Bonferroni critical value for twelve one-sided tests; and for the neural network from 1.82 (EM-trained) to 3.88 (DM-trained). The two statistics answer different questions. The Clark–West test asks whether the forecast deviations from the benchmark are informative, correcting for the noise that estimating them adds [Clark and West, 2007]; R_{OS}^2 asks whether the forecasts, as issued, reduce squared error. The slope column reconciles them. Regressing the realized deviation $r - \tilde{r}$ on the forecast deviation $\hat{r} - \tilde{r}$ gives slopes of 0.19 to 0.40, so the forecasts point in the right direction but are two and a half to five times too large, and the squared error they add exceeds the signal they carry. The EM-trained neural network’s forecasts are the most overstated (slope 0.19). Because the slope is estimated over the whole out-of-sample period, it is a diagnostic, not a rescaling an investor could have applied in real time.

Table 2: Out-of-sample R^2 for emerging-market factor returns, 2000–2025

Model	Training	vs. own historical mean		vs. shrinkage mean		Slope	Obs.
		R_{OS}^2 (%)	CW t	R_{OS}^2 (%)	CW t		
OLS	EM	1.148	8.72	−0.055	1.61	0.35	801,536
	DM only	1.011	8.70	−0.194	2.10	0.29	801,536
	Pooled	1.108	8.82	−0.096	2.10	0.35	801,536
Elastic net	EM	1.157	8.73	−0.046	1.56	0.36	801,536
	DM only	1.031	8.69	−0.174	2.08	0.30	801,536
	Pooled	1.119	8.83	−0.084	2.03	0.35	801,536
Gradient-boosted trees	EM	0.993	8.88	−0.212	3.66	0.30	801,536
	DM only	0.973	8.45	−0.233	3.79	0.35	801,536
	Pooled	1.096	8.95	−0.108	4.25	0.40	801,536
Neural network	EM	0.832	8.45	−0.376	1.82	0.19	801,536
	DM only	0.728	8.33	−0.481	3.88	0.27	801,536
	Pooled	0.910	8.65	−0.296	3.08	0.29	801,536
Factor momentum	–	−6.169	5.01	−7.461	2.41	–	801,536
Shrinkage mean	–	1.203	9.05	0.000	–	–	801,536

Notes: Pooled over all emerging-market country-factor-months. $R_{OS}^2 = 1 - \sum(r - \hat{r})^2 / \sum(r - b)^2$ for benchmark b . CW t is the Clark–West statistic: the adjusted MSPE difference is averaged across the cross-section each month and its mean tested with Newey–West standard errors (3 lags); one-sided 5% critical value 1.645. Training scope: EM = emerging markets only; DM only = developed markets only (the transfer test); Pooled = both. Learned models forecast the deviation from the shrinkage mean and are refitted every January. The shrinkage mean is the benchmark itself, so its own row against that benchmark is zero by construction. Slope: pooled regression slope, without intercept, of $r - \tilde{r}$ on $\hat{r} - \tilde{r}$ over the whole out-of-sample period; a descriptive diagnostic that uses future data and is not a tradable rescaling.

5.2 Does developed-market training transfer?

The transfer test holds the emerging-market test observations fixed and changes only the training sample. Models trained only on developed markets have lower R_{OS}^2 against the shrinkage mean than models trained on emerging markets (elastic net: -0.17% vs. -0.05% ; trees: -0.23% vs. -0.21% ; neural network: -0.48% vs. -0.38%) but Clark–West statistics at least as large (2.08 vs. 1.56; 3.79 vs. 3.66; 3.88 vs. 1.82). Their forecasts are also more dispersed: the standard deviation of the forecast deviation from the shrinkage mean is 0.29% a month for the DM-trained elastic net and 0.18% for the EM-trained one. Developed-market history thus carries directional information about emerging-market factor returns of about the same strength as emerging-market history, but it produces larger forecasts that are penalized in squared error. Training on both samples gives the largest tree statistic (4.25) and the least negative tree R_{OS}^2 (-0.11%). Across the fourteen elastic-net and tree comparisons in the robustness specifications of Table 5, the DM-trained statistic is the larger in ten; it is smaller by more than 0.1 only for trees evaluated from 2005 (3.11 vs. 3.51) and under the time-varying classification (3.28 vs. 3.66). These are out-of-sample predictive comparisons. They do not identify why developed-market history is informative — common risk, common investor behavior or global capital flows.

Table 3 breaks the comparison down by market for the elastic net, the learned model with the highest pooled R_{OS}^2 against the shrinkage mean. EM training gives a positive R_{OS}^2 in 12 of 22 markets and a Clark–West statistic above 1.645 in 2; DM training gives 8 and 5. The country-level estimates are noisy and not adjusted for multiple testing, so I read them as showing that neither training sample dominates market by market, not as identifying markets where transfer works.

5.3 Economic value

Table 4 and Figure 1 report the timing portfolios. The static portfolio, which holds every available factor equally within each market, earns 2.42% a year with volatility of 1.60%, a gross Sharpe ratio of 1.51. Learned-model portfolios earn more before costs, 3.06–3.83% a year, and the tree portfolios have higher gross Sharpe ratios (1.82–1.94). But they trade heavily: monthly turnover is 0.41–0.63, against 0.02 for the static portfolio. At 20 basis points per unit of turnover their net Sharpe ratios fall to 0.71–1.26, all below the static portfolio’s 1.48. The factor-momentum rule does worse (0.77 net).

The highest net Sharpe ratio, 1.91 (2.00 gross), belongs to the portfolio that weights factors by the shrinkage mean. That portfolio forecasts no time variation: it tilts toward factors whose pooled historical premia are high and rebalances little (turnover 0.06). Weighting by the unshrunk own mean instead gives 1.41. In this design, then, the economic gain in emerging markets comes from a better estimate of which factors earn a premium, not from timing when they do.

Two features of the data raise the level of every Sharpe ratio in Table 4, which should therefore be read comparatively. The factor returns are gross of the costs of their own monthly rebalancing,

Table 3: Transfer by country: elastic net trained on emerging vs. developed markets

Market	EM-trained		DM-trained		Obs.
	R^2_{OS} (%)	CW t	R^2_{OS} (%)	CW t	
Brazil	0.038	−0.49	0.168	0.45	35,480
Chile	−0.349	−1.35	−0.688	−1.18	42,971
China	0.274	1.72	−0.158	0.67	43,438
Colombia	0.107	1.14	0.105	0.98	22,468
Egypt	−0.597	−1.70	−1.102	−1.06	20,216
Greece	−0.387	−0.79	−0.475	−0.10	33,497
India	0.370	1.34	0.567	2.45	41,679
Indonesia	0.018	1.03	0.198	1.78	42,017
Korea	−0.341	−0.44	−0.831	−1.18	44,157
Kuwait	0.003	0.73	0.040	1.29	29,444
Malaysia	0.220	1.30	0.295	2.15	45,139
Mexico	−0.052	0.68	−0.086	1.35	44,203
Peru	0.001	0.79	−0.227	0.50	25,338
Philippines	−0.378	−1.16	−0.775	−1.06	35,553
Poland	0.165	0.99	0.502	2.23	36,832
Qatar	0.004	1.46	−0.340	1.15	24,445
Saudi Arabia	0.263	0.93	−0.102	0.73	34,423
South Africa	−0.121	0.35	−0.499	−0.41	45,986
Taiwan	−0.023	0.39	−0.166	0.56	42,731
Thailand	−0.170	−0.05	−0.168	0.96	43,196
Turkey	−0.007	0.95	−0.254	0.57	42,183
United Arab Emirates	0.164	2.35	0.161	1.86	26,140
Markets with $R^2_{OS} > 0$	12 of 22		8 of 22		
Markets with CW $t > 1.645$	2 of 22		5 of 22		

Notes: R^2_{OS} and Clark–West t against the shrinkage mean, computed within each market on the same observations for both training scopes. Country-level tests are not adjusted for multiple comparisons.

which are larger in emerging markets, and each factor’s sign is fixed by the original study that proposed it, several of which were published after 2000.

5.4 Robustness

Table 5 varies the benchmark’s prior weight κ , the start of the evaluation sample, the weighting of the factor portfolios, the market classification and the cost. The time-varying classification assigns each country-month the MSCI class then in force (Appendix B): it moves Greece out of the emerging-market sample for 2001–2013, adds Argentina, Morocco, Pakistan and Russia during their emerging-market years, and admits Qatar, the United Arab Emirates, Saudi Arabia and Kuwait only after their upgrades. The neural network is run only in the baseline.

Three results are stable. In every row the shrinkage mean beats the own historical mean, by 0.97–1.37% in R^2_{OS} , and the shrinkage-mean portfolio has the highest net Sharpe ratio. And the Clark–West statistics of the EM- and DM-trained trees stay above 2.8, with one exception: EM-trained trees on uncapped value-weighted factors (1.21).

Table 4: Emerging-market factor-timing portfolios

Strategy	Training	Mean (%)	Vol. (%)	SR gross	SR net	Turnover
Static equal-weight	–	2.42	1.60	1.51	1.48	0.02
OLS	EM	3.08	1.91	1.61	1.07	0.43
	DM only	3.07	2.19	1.40	0.71	0.63
	Pooled	3.14	2.10	1.50	0.86	0.56
Elastic net	EM	3.10	1.88	1.64	1.11	0.41
	DM only	3.10	2.18	1.42	0.74	0.62
	Pooled	3.16	2.09	1.51	0.88	0.54
Gradient-boosted trees	EM	3.34	1.72	1.94	1.26	0.48
	DM only	3.83	2.10	1.82	1.13	0.61
	Pooled	3.69	2.01	1.83	1.17	0.55
Neural network	EM	3.06	1.86	1.65	1.07	0.45
	DM only	3.43	2.02	1.70	0.99	0.60
	Pooled	3.23	2.09	1.54	0.90	0.56
Shrinkage mean	–	3.19	1.60	2.00	1.91	0.06
Own historical mean	–	2.32	1.51	1.53	1.41	0.07
Factor momentum (sign)	–	2.02	1.84	1.10	0.77	0.25

Notes: Within each emerging market and month, the timing portfolio holds each factor in proportion to its forecast, scaled to unit gross exposure; the static portfolio holds every available factor equally. Country portfolios are averaged with equal weights. Mean and volatility are annualized, gross of timing costs. Net Sharpe ratios deduct 20 basis points per unit of monthly turnover; factor returns are already gross of their own rebalancing costs. The factor-momentum rule holds each factor long or short by the sign of its trailing 12-month mean.

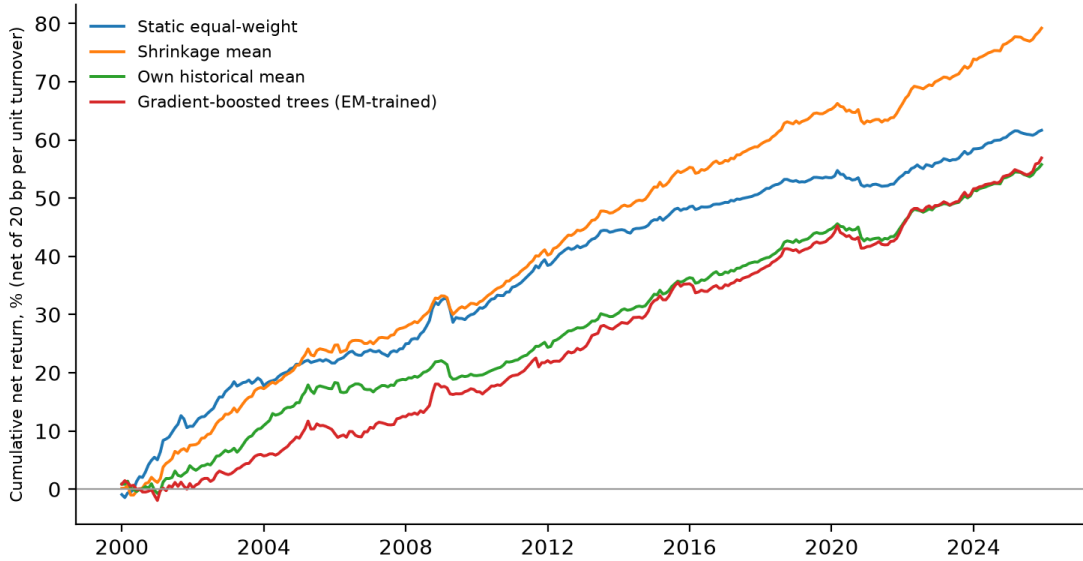


Figure 1: Cumulative net returns of emerging-market factor-timing portfolios, 2000–2025. Sum of monthly returns net of 20 basis points per unit of turnover, for the static equal-weight portfolio and the three strategies with the highest net Sharpe ratios.

Three results depend on the specification. First, the linear models’ Clark–West statistics are sensitive to κ and to factor weighting: 3.1–3.3 with $\kappa = 60$, but 0.8–1.5 with $\kappa = 240$ and 1.3–1.5 with uncapped value-weighted factors. A weaker prior leaves more of the noise in each country-factor mean in the benchmark, which the models can again correct by pooling — the same mechanism that inflates tests against the own mean. The linear-model evidence of timing on value-weighted factors is therefore fragile.

Second, equal-weighted factors behave differently. Seven of the nine learned model-by-scope combinations beat the shrinkage mean, with R_{OS}^2 up to 0.17% (pooled elastic net) and Clark–West statistics of 3.8–5.0, and every learned-model portfolio beats the static portfolio net of 20 basis points (2.28–3.08 against 2.01), though none beats the shrinkage-mean portfolio (3.90). Equal weighting gives more weight to small stocks, which are the most expensive to trade, and the cost charged per unit of turnover is the same for every weighting, so these Sharpe ratios are the least implementable in the table.

Third, the comparison of timing portfolios with the static portfolio depends on costs: at 10 basis points the EM-trained tree portfolio’s net Sharpe ratio (1.60) exceeds the static portfolio’s (1.49), though it stays below the shrinkage-mean portfolio’s (1.95); at 40 basis points the learned-model portfolios fall to 0.05–0.59. Table 6 reports the cost at which each timing portfolio’s net Sharpe ratio equals the static portfolio’s. On value-weighted factors it is at most 16 basis points per unit of turnover, and the DM-trained elastic net never beats the static portfolio; on equal-weighted factors it is 30–48 basis points. Whether costs that low are attainable for the small emerging-market stocks that dominate equal-weighted factors is a question this paper does not answer. Stronger shrinkage also improves the benchmark itself: its R_{OS}^2 against the own mean rises from 0.98% at $\kappa = 60$ to 1.36% at $\kappa = 240$, which suggests that emerging-market factor premia differ across countries less than their sample means do.

Not yet run: trading costs that vary with stock size.

5.5 Exploratory extensions

Two extensions were chosen after seeing the results above, so they are exploratory rather than tests specified in advance. Table 7 reports both.

Real-time forecast shrinkage. The slope diagnostic in Table 2 suggests scaling forecasts toward the benchmark, but that slope uses the whole out-of-sample period. Here the scale is estimated each January inside the training window only: the model is fitted on labels before the window’s last 36 months, the slope of realized on forecast deviations over those months (clipped to $[0, 1]$) is the scale, and the model is refitted on the whole window. Average yearly scales are 0.41–0.66. On capped value-weighted factors, shrinkage raises every model’s R_{OS}^2 against the shrinkage mean, but none becomes positive; the best is -0.02% (EM-trained trees). It cuts turnover by a third to a half, and the EM-trained portfolios’ net Sharpe ratios rise to 1.51–1.71, above the static portfolio’s 1.48

Table 5: Robustness

Panel A: R_{OS}^2 (%) and Clark–West t against the shrinkage mean

Specification	Shrink. vs. own mean	Elastic net, EM		Elastic net, DM		GBRT, EM		GBRT, DM	
		R^2	t	R^2	t	R^2	t	R^2	t
Baseline	1.203	−0.046	1.56	−0.174	2.08	−0.212	3.66	−0.233	3.79
OOS from 2005	0.972	−0.005	1.89	−0.132	1.99	0.011	3.51	−0.170	3.11
$\kappa = 60$	0.976	−0.004	3.13	−0.129	3.08	−0.195	4.48	−0.176	4.46
$\kappa = 240$	1.356	−0.069	0.79	−0.197	1.52	−0.265	2.88	−0.261	3.45
Equal-weighted factors	1.370	0.114	3.79	0.103	4.14	−0.020	4.34	−0.041	4.74
Uncapped value-weighted	1.194	−0.054	1.28	−0.129	1.37	−0.294	1.21	−0.247	3.14
Time-varying classification	1.229	−0.047	1.84	−0.183	2.02	−0.240	3.66	−0.318	3.28

Panel B: net Sharpe ratios of timing portfolios

Specification	Static	Shrinkage mean	Elastic net		GBRT	
			EM	DM	EM	DM
Baseline	1.48	1.91	1.11	0.74	1.26	1.13
OOS from 2005	1.34	1.79	1.25	0.77	1.34	0.99
$\kappa = 60$	1.48	1.73	1.00	0.70	1.15	1.09
$\kappa = 240$	1.48	2.05	1.19	0.77	1.31	1.15
Equal-weighted factors	2.01	3.90	2.76	2.31	3.08	2.51
Uncapped value-weighted	1.45	1.80	1.03	0.74	1.10	1.06
Time-varying classification	1.56	1.97	1.19	0.83	1.43	1.05
Cost 10 bp	1.49	1.95	1.38	1.08	1.60	1.47
Cost 40 bp	1.45	1.81	0.58	0.05	0.59	0.43

Notes: Emerging-market test set. Baseline: $\kappa = 120$, 20 bp per unit of turnover, out of sample 2000–2025. “OOS from 2005” keeps the baseline forecasts for target months from 2005, which equal those of a rerun starting in 2005 because models are refitted annually on an expanding window. κ rows rerun the full pipeline, changing both the benchmark and the learned models’ anchor. Factor-weighting rows rerun it on equal-weighted or uncapped value-weighted JKP factors. “Time-varying classification” assigns each country-month the MSCI class then in force (Appendix B). Cost rows change only the charge per unit of timing turnover. “Shrink. vs. own mean” is the R_{OS}^2 of the shrinkage mean against the own historical mean. EM / DM = trained on emerging / developed markets only.

Table 6: Break-even trading cost of timing portfolios (basis points per unit of turnover)

Specification	Elastic net, EM	Elastic net, DM	GBRT, EM	GBRT, DM
Baseline	6	never	13	10
Equal-weighted factors	44	30	47	36
Uncapped value-weighted	3	never	8	6
Time-varying classification	5	never	16	4

Notes: Cost per unit of monthly timing turnover at which the timing portfolio’s net Sharpe ratio equals that of the static equal-weight factor portfolio, which is charged the same cost on its own turnover. “never”: the timing portfolio’s Sharpe ratio is lower even at zero cost. EM / DM = trained on emerging / developed markets only. Against the shrinkage-mean portfolio the break-even cost is positive only for GBRT, EM (time-varying classification, 1 bp).

but below the shrinkage-mean portfolio’s 1.91. The Clark–West statistics of the EM-trained linear models fall to 0.8–0.9, because in several years the estimated scale is zero and the forecast deviation disappears. On equal-weighted factors, shrinkage turns the EM-trained trees’ R_{OS}^2 positive (0.13%)

and lifts their portfolio to 3.65, still below the shrinkage-mean portfolio’s 3.90.

U.S. state variables. A developed-to-emerging channel that runs through global risk or funding conditions suggests conditioning on U.S. state variables. Adding log VIX, the 10-year minus 3-month Treasury spread and the 12-month change of the broad dollar as predictors makes every model worse on the same sample (Panel B, which starts in 1990 because VIX does). The linear models lose 0.04–0.07 percentage points of R_{OS}^2 . The trees deteriorate most: their R_{OS}^2 falls to -0.38% (EM-trained) and -1.05% (DM-trained), and their Clark–West statistics to 1.37 and 0.58. Because the state variables are common to every series in a month, one plausible reading is that the trees use them to fit month-specific effects that do not recur.

6 Conclusion

Emerging-market factor returns contain a small amount of predictable time variation. It is detectable out of sample, including by models trained only on developed markets, but in capped value-weighted factors it is too small relative to forecast noise and trading costs to be worth timing. In equal-weighted factors, which lean toward small stocks, it is larger, and timing portfolios beat a static factor portfolio net of a uniform 20-basis-point cost that likely understates the cost of trading those stocks. The larger and more robust gain is in estimating the unconditional premia themselves: pooling each factor’s history across emerging markets beats the own historical mean with an R_{OS}^2 of 1.20% and gives the best portfolio net of costs in every specification. The methodological point follows. In multi-country factor panels, out-of-sample tests against the own historical mean mostly measure that pooling gain, and a pooled benchmark should replace it.

Shrinking forecasts toward the benchmark in real time narrows the gap to the shrinkage-mean tilt without closing it, and U.S. state variables do not help. Two directions remain: trading costs that vary with stock size, which would test whether the equal-weighted gains survive, and state variables that vary across emerging markets rather than U.S. aggregates, which would say more about why developed-market history is informative.

Table 7: Exploratory extensions

Model	Forecast	EM-trained			DM-trained		
		R_{OS}^2 (%)	CW t	SR net	R_{OS}^2 (%)	CW t	SR net
<i>Panel A: real-time forecast shrinkage, out of sample 2000–2025</i>							
Capped value-weighted factors							
OLS	As estimated	−0.055	1.61	1.07	−0.194	2.10	0.71
	Real-time shrinkage	−0.031	0.82	1.51	−0.139	1.39	1.02
Elastic net	As estimated	−0.046	1.56	1.11	−0.174	2.08	0.74
	Real-time shrinkage	−0.023	0.90	1.53	−0.129	1.39	1.04
GBRT	As estimated	−0.212	3.66	1.26	−0.233	3.79	1.13
	Real-time shrinkage	−0.019	2.04	1.70	−0.131	2.52	1.48
Static portfolio				1.48			
Shrinkage-mean portfolio				1.91			
Equal-weighted factors							
OLS	As estimated	0.100	3.76	2.69	0.076	4.12	2.28
	Real-time shrinkage	0.120	3.68	3.32	0.076	3.16	2.62
Elastic net	As estimated	0.114	3.79	2.76	0.103	4.14	2.31
	Real-time shrinkage	0.128	3.78	3.35	0.096	3.22	2.64
GBRT	As estimated	−0.020	4.34	3.08	−0.041	4.74	2.51
	Real-time shrinkage	0.125	4.31	3.65	0.076	3.92	2.91
Static portfolio				2.01			
Shrinkage-mean portfolio				3.90			
<i>Panel B: U.S. state variables, feature months from 1990, out of sample 2001–2025</i>							
OLS	Without state variables	−0.023	1.98	1.21	−0.212	2.16	0.79
	With state variables	−0.088	1.47	1.05	−0.264	2.00	0.68
Elastic net	Without state variables	−0.016	1.94	1.25	−0.191	2.13	0.81
	With state variables	−0.069	1.47	1.11	−0.234	1.98	0.72
GBRT	Without state variables	−0.208	3.25	1.34	−0.323	3.38	1.11
	With state variables	−0.383	1.37	0.85	−1.051	0.58	0.43
Static portfolio				1.44			
Shrinkage-mean portfolio				1.99			

Notes: R_{OS}^2 and Clark–West t against the shrinkage mean; SR net is the net Sharpe ratio of the timing portfolio at 20 basis points per unit of turnover (for the static and shrinkage-mean rows, of those portfolios). Real-time shrinkage multiplies each forecast’s deviation from the shrinkage mean by a slope in $[0, 1]$ estimated inside the training window: the model is fitted on labels before the window’s last 36 months and the slope of realized on forecast deviations over those months is the scale; the model is then refitted on the whole window. Average yearly scales range from 0.41 to 0.66. State variables: log VIX, the U.S. 10-year minus 3-month Treasury spread and the 12-month log change of the broad trade-weighted dollar (FRED), dated at the end of the feature month; both rows of Panel B use the same sample. Both extensions were chosen after seeing the baseline results and are exploratory.

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A Validation on known-truth simulated panels

The pipeline is validated on simulated panels in which the predictable part of returns is known. For country c and factor f ,

$$r_{c,f,t+1} = \mu_{c,f} + x_{c,f,t} + e_{c,f,t+1}, \quad x_{c,f,t} = \sqrt{\rho} z_{f,t} + \sqrt{1-\rho} u_{c,f,t}, \quad (3)$$

where $\mu_{c,f} \sim N(0.002, 0.003^2)$, z and u are independent AR(1) processes with persistence 0.95 and standard deviation σ_x , $\rho = 0.6$ is the share of the predictable component common to all countries, and $e \sim N(0, 0.03^2)$. Setting $\sigma_x = 0$ removes all predictability.

Why the own mean over-rejects. The Clark–West term $2(r-b)(\hat{r}-b)$ has positive expectation whenever the forecast corrects the benchmark’s own estimation error. With heterogeneous $\mu_{c,f}$ and short histories, $\bar{r}_{i,t}$ is noisy, and a pooled model learns to shrink it — a genuine improvement in estimating $\mu_{c,f}$ that the test attributes to timing. Anchoring on the shrinkage mean (1) removes most of that error before the model is fitted.

Table 8: Size and power of the Clark–West test on known-truth synthetic panels

Signal s.d.	Draws	vs. own historical mean		vs. shrinkage mean	
		Mean R_{OS}^2 (%)	Reject (%)	Mean R_{OS}^2 (%)	Reject (%)
0 (null)	40	0.058	87.5	-0.055	5.0
0.004	40	0.264	97.5	0.133	92.5
0.008	40	1.773	100.0	1.599	100.0

Notes: Simulated data, not a finding. Each draw is a panel of 6 developed and 6 emerging markets, 12 factors and 300 months generated by equation (3); the pooled OLS model is evaluated out of sample from 2005. “Reject” is the share of draws with a one-sided Clark–West t -statistic above 1.645. The first row has no predictability, so its rejection rate is the empirical size of a nominal 5% test.

Table 8 reports 40 draws at each signal level. Against the own historical mean, the test rejects in 87.5% of null draws. Against the shrinkage mean, the rejection rate under the null is 5.0%, equal to the nominal size, and power is 92.5% and 100% at the two signal levels. In separate draws with predictability shared across countries ($\rho = 0.6$), models trained only on developed markets also recover it in emerging markets, so the transfer test can detect transfer when it exists.

B Time-varying market classification

The baseline classifies markets by the current MSCI classification. The robustness check in Table 5 instead uses the class in force in each month. Table 9 lists the reclassifications within the sample that involve developed or emerging status; each takes effect from the first full month after MSCI implemented it. A market’s training and test assignment use its class in month t . The pooled regional mean for region g averages every past country-month that belonged to g when it was observed, and the developed-market predictor averages the markets classified as developed at t . First inclusion in the MSCI indexes and partial-inclusion steps are not modelled.

Table 9: MSCI reclassifications used for the time-varying classification

Market	In force from	New class	MSCI event
Portugal	1997-12	Developed	Emerging → Developed, Nov 1997
Greece	2001-06	Developed	Emerging → Developed, May 2001
Venezuela	2006-06	Frontier/standalone	Emerging → Standalone, May 2006
Jordan	2008-12	Frontier/standalone	Emerging → Frontier, Nov 2008
Pakistan	2009-01	Frontier/standalone	Emerging → Standalone, close of 31 Dec 2008
Argentina	2009-06	Frontier/standalone	Emerging → Frontier, May 2009
Israel	2010-06	Developed	Emerging → Developed, May 2010
Greece	2013-12	Emerging	Developed → Emerging, Nov 2013
Morocco	2013-12	Frontier/standalone	Emerging → Frontier, Nov 2013
United Arab Emirates	2014-06	Emerging	Frontier → Emerging, May 2014
Qatar	2014-06	Emerging	Frontier → Emerging, May 2014
Pakistan	2017-06	Emerging	Frontier → Emerging, effective 1 Jun 2017
Argentina	2019-06	Emerging	Frontier → Emerging, May 2019
Saudi Arabia	2019-06	Emerging	Standalone → Emerging, May 2019 (step 1 of 2)
Kuwait	2020-12	Emerging	Frontier → Emerging, close of 30 Nov 2020
Argentina	2021-12	Frontier/standalone	Emerging → Standalone, Nov 2021
Pakistan	2021-12	Frontier/standalone	Emerging → Frontier, Nov 2021
Russia	2022-04	Frontier/standalone	Emerging → Standalone, close of 9 Mar 2022

Notes: Each market takes the new class from the first full month after MSCI implemented the change; before its first event a market has its prior class, and markets without events keep their current class. First inclusion in the MSCI indexes and partial-inclusion steps are not modelled. Sources: MSCI’s list of market reclassifications and MSCI announcements, cited in `config/market_history.yaml`.